

HAMZA SIMOU

Software Engineering Student | Full-Stack & DevOps
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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Final-year Software Engineering student at EMSI focused on backend, DevOps, and CI/CD, with hands-on experience building production full-stack and AI-powered systems. Seeking a PFE internship for 2027.

EDUCATION

State Engineering Diploma — Software Engineering 2022 – 2027
EMSI Marrakech (Bac+5)

TECHNICAL SKILLS

- **Languages:** Java (17/21), Python, JavaScript, TypeScript
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, REST APIs, Microservices, Node.js (Express.js)
- **Frontend & Mobile:** Angular 21, Flutter, React, Next.js
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQL Server
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins, AWS, GitHub Actions, CI/CD, Linux
- **Security:** Keycloak, OAuth2, JWT
- **Methodologies:** Agile/Scrum, UML, OOP, Design Patterns
- **Tools:** Git, GitHub, GitLab, Maven, Jira, Nexus

EXPERIENCE

Software Engineering Intern — ERP Systems (Odoo 17) June 2026 – September 2026
AMANUS, Marrakech, Morocco | *On-site* | *GitHub*

- Assisted in developing custom Odoo 17 modules in Python for restaurant ERP (Sales, Inventory, Procurement).
- Built and modified 10+ XML views and added custom fields to existing models using the Odoo ORM.
- Fixed ~20 bug tickets and tested features in a Docker-based development environment.
- Collaborated with a team of 5 developers using Git and GitHub for version control.
- Technologies: Odoo 17, Python, PostgreSQL, XML, Docker, GitHub Actions

Software Engineering Intern July 2025 – August 2025
Sofitel Marrakech, IT Department | *On-site* | *GitHub*

- Designed and built Hotel Asset Recovery, a lost-and-found platform that digitized the hotel's paper process, cutting item lookup from ~15–20 minutes to under a minute.
- Built a modular Spring Boot backend (Spring Modulith) with Kafka events, Keycloak OAuth2/OIDC security, and MinIO photo storage.
- Built a hybrid AI matching engine (Ollama vision + text embeddings, pgvector) combining visual, semantic, and location signals, with duty-manager confirmation before return.
- Technologies: Java 21, Spring Boot 4, Spring Security, Spring Data JPA, Angular 21, Keycloak, PostgreSQL, Kafka, MinIO, Ollama, Docker Compose, Prometheus, Grafana, Tempo

PROJECTS

Parkingo — Smart Parking Platform (1st Prize EMSIstes INNOV'26)
Spring Boot, Flutter, Next.js, MySQL, Gemini AI, Mapbox | *GitHub*

- Built an end-to-end smart parking system (Spring Boot, Next.js, Flutter, MySQL) for reservation, payment, maps, and gate control.
- Implemented JWT auth, wallet payments, time-conflict slot booking, and scheduled slot release.
- Added Gemini conversational booking, license-plate OCR (EasyOCR), QR validation, and a 3D parking twin.
- Supported four roles (driver, owner, guard, admin) with Mapbox maps and FR/EN/AR.

KickPro — AI Powered Football Talent Platform
Spring Boot, Flutter, Kafka, Gemini AI, AWS | *GitHub*

- Full-stack Flutter + Spring Boot platform (46 screens, 127 secured REST APIs, JWT/RBAC) for player–scout discovery.
- Pessimistic locking + overlap checks for stadium match booking; Kafka-driven booking notifications; WebSocket chat.
- Gemini-powered scout assist and AI coach; Cloudinary uploads; OpenStreetMap in the mobile app.
- GitHub Actions CI/CD, Docker, AWS (EC2/RDS), 32 JUnit tests, Angular marketing site.

LANGUAGES

Arabic (Native) · French (C1) · English (C1)